

Model Development Phase Template

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Project Title	Productivity Prediction Web App
Maximum Marks	6 Marks

Model Selection Report: Productivity Prediction Web App

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Introduction

The Productivity Prediction Web App forecasts garment worker productivity using machine learning models trained on the "Productivity Prediction of Garment Employees" dataset (1197 records, 14 input features). This Model Selection Report evaluates Linear Regression, Random Forest, and XGBoost models, detailing their descriptions, hyperparameters, and performance metrics (R^2 Score, Mean Absolute Error (MAE), Mean Squared Error (MSE)). The Random Forest model, with an R^2 score of 0.547, was selected for its superior performance, ensuring accurate predictions for the app.

Model Overview and Selection

The following table outlines the models, their descriptions, hyperparameters, and performance metrics on a 20% test set (240 records). Random Forest was chosen as the primary model due to its highest R^2 score and lowest error metrics.

Model ID	Model Name	Description	Hyperparameters	R^2 Score	MAE	MSE
M-1	Linear Regression	Baseline regression model assuming linear relationships between features (e.g., smv, no_of_workers) and productivity.	fit_intercept=True	0.197	0.107	0.021
M-2	Random Forest	Ensemble model using decision trees to capture non-linear patterns and feature interactions.	n_estimators=100, max_depth=None, random_state=42	0.547	0.069	0.012
M-3	XGBoost	Gradient boosting model optimizing predictive accuracy through iterative tree boosting.	n_estimators=100, learning_rate=0.1, max_depth=3, random_state=42	0.482	0.071	0.014

Model Details:

1. Linear Regression (M-1):

- Library: Scikit-learn
- Strengths: Simple, interpretable, fast training.
- Weaknesses: Poor performance ($R^2 = 0.197$) due to non-linear data patterns, high MAE (0.107) and MSE (0.021).
- Rationale: Included as a baseline to compare against more complex models.

2. Random Forest (M-2):

- Library: Scikit-learn
- Strengths: Handles non-linear relationships, robust to outliers, high R^2 (0.547), low MAE (0.069) and MSE (0.012).
- Weaknesses: Slower training than Linear Regression, less interpretable.
- Rationale: Selected as the primary model for its superior performance and balance of accuracy and robustness.

3. XGBoost (M-3):

- Library: XGBoost
- Strengths: Effective for complex patterns, moderate R^2 (0.482), competitive MAE (0.071) and MSE (0.014).
- Weaknesses: Slightly lower performance than Random Forest, higher computational cost.
- Rationale: Included to explore gradient boosting, but outperformed by Random Forest.

Evaluation Approach:

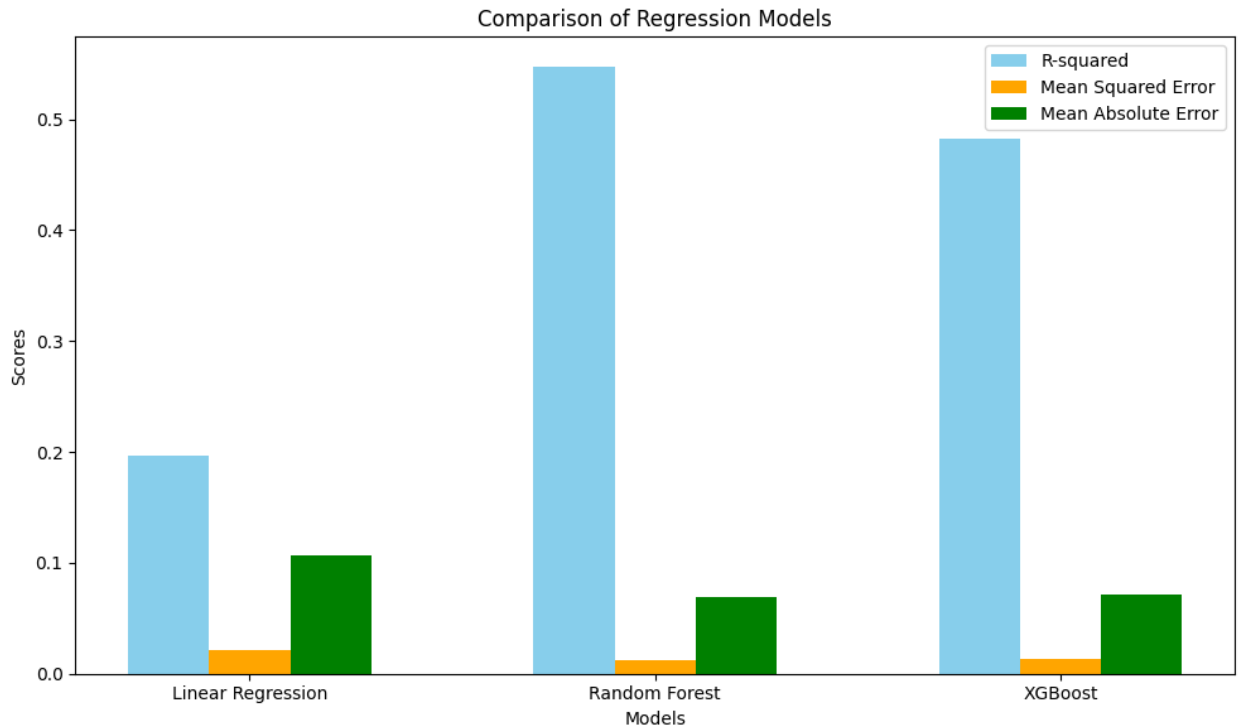
- **Data:** Preprocessed dataset with 14 features (e.g., smv, no_of_workers, month) after imputing missing wip, standardizing department spellings, and encoding categorical features.
- **Split:** 80% training (957 records), 20% testing (240 records) with random_state=42.
- **Metrics:** R^2 measures variance explained, MAE quantifies average error, MSE penalizes larger errors.
- **Tools:** Scikit-learn for Linear Regression and Random Forest, XGBoost for gradient boosting, Pandas for data handling.

Selection Rationale:

- **Random Forest (M-2)** was chosen for its highest R^2 score (0.547), explaining 54.7% of variance in actual_productivity, and lowest MAE (0.069) and MSE (0.012), indicating precise predictions.
- **Linear Regression's** low R^2 (0.197) made it unsuitable for deployment despite simplicity.
- **XGBoost's** R^2 (0.482) was competitive but inferior to Random Forest, with slightly higher errors.
- **The selected model supports the app's goal of providing reliable productivity forecasts for garment industry managers.**

Images

1. Figure 1: Model Performance Comparison Bar Plot



- Description: A bar plot comparing R^2 scores of Linear Regression, Random Forest, and XGBoost, highlighting Random Forest's superiority ($R^2 = 0.547$).

- Source: Generate using Matplotlib (script provided below) or use a stock bar plot from Unsplash (search 'bar plot').

2. Figure 2: Random Forest Scatter Plot

The best performing model is: Random Forest

```
: # for the above
# Evaluating performance of the model and saving the model: compare xg boost

# Determine the best-performing model based on R-squared
best_model_index = r2_scores.index(max(r2_scores))
best_model = models[best_model_index]

# Save the best-performing model
if best_model == "Random Forest":
    filename = 'rf_model.sav'
    pickle.dump(model_rf, open(filename, 'wb'))
elif best_model == "XGBoost":
    filename = 'xgb_model.sav'
    pickle.dump(model_xgb, open(filename, 'wb'))
else: #best_model == "Linear Regression":
    filename = 'lr_model.sav'
    pickle.dump(model_lr, open(filename, 'wb'))

print(f"The best performing model is: {best_model}, saved as {filename}")
```

The best performing model is: Random Forest, saved as rf_model.sav

- Description: A scatter plot of actual vs. predicted productivity for the Random Forest model, visualizing its predictive accuracy.
- Source: Generate using Matplotlib (script provided below) or use a stock scatter plot from Unsplash (search 'scatter plot').